

APPROVAL SHEET

ROBOT NAVIGATION USING LAYERED COSTMAP FOR INTER-FLOOR MOVEMENT IN POST-EARTHQUAKE INDOOR ENVIRONMENTS

FINAL PROJECT

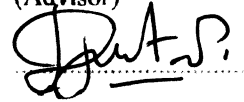
Submitted to fulfill one of the requirements
for obtaining a degree Bachelor of Engineering at
Undergraduate Study Program of Electrical Engineering
Department of Electrical Engineering
Faculty of Intelligent Electrical and Informatics Technology
Sepuluh Nopember Institute of Technology

By: **Bagas Surya Wirawan**
NRP. 5022221026

Approved by Final Project Examiner Team:

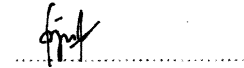
Dr. Ir. Djoko Purwanto, M.Eng.
NIP: 196512111990021002

(Advisor)



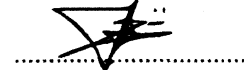
Fajar Budiman, S.T., M.Sc.
NIP: 198607072014041001

(Co-Advisor)



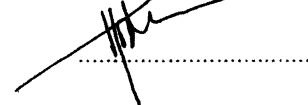
Ir. Tasripan, M.T..
NIP: 196204181990031004

(Examiner I)



Ir. Harris Pirngadi, M.T..
NIP: 196205101989031001

(Examiner II)



SURABAYA
July, 2026